

Real-Time Pakistan Sign Language Recognition using ST-GCN

1. Introduction & Choice of Model

Sign language recognition is a complex task that requires capturing both the static "shape" of a hand and the "motion" of the gesture over time. For this project, we chose the **Spatial-Temporal Graph Convolutional Network (ST-GCN)**. Unlike traditional models that treat video as a flat sequence of images, the ST-GCN treats the human body as a dynamic graph. This allows the model to focus purely on the movement of joints (skeleton data), making it far more efficient and robust than pixel-based models when dealing with varied backgrounds or lighting conditions.

2. Tools and Technologies

The development of this model utilized a high-performance software stack tailored for deep learning:

- **MediaPipe:** Used for high-fidelity landmark extraction (Pose, Hands, and Face).
- **TensorFlow & Keras:** The core frameworks for building and training the deep neural layers.
- **Python (NumPy/Pandas):** Essential for handling the massive dataset of **422,073 coordinate files**.
- **Scikit-learn:** Utilized for rigorous **80/20 train-test splitting** to ensure the model generalizes to new users.
- **Matplotlib/Seaborn:** Used for generating real-time performance dashboards and confusion matrices.

3. Architecture of the ST-GCN

The architecture is divided into two primary dimensions:

- **Spatial Dimension:** The model uses an **Adjacency Matrix** to define the physical connections of the 75-node skeleton (e.g., the "elbow" is connected to the "wrist"). Spatial Graph Convolutions learn the geometric patterns of specific signs.
- **Temporal Dimension:** 1D convolutions are applied across the time axis (30 frames per video) to capture the "rhythm" and "velocity" of the gestures.
- **Global Average Pooling:** This layer condenses the complex graph features into a final vector for the **Softmax** classifier, which predicts one of the **82 PSL signs**.

4. Why This Model? (Comparative Analysis)

We chose ST-GCN over other popular architectures for several strategic reasons:

- **vs. CNN (Convolutional Neural Networks):** CNNs are computationally expensive as they process every pixel. ST-GCN only processes 75 points per frame, allowing us to train on **400k+ samples** even on an 8GB RAM system.
- **vs. LSTM (Long Short-Term Memory):** While LSTMs are good for sequences, they often struggle to capture the complex spatial "shape" of a hand sign. ST-SLN excels by combining spatial geometry with temporal sequence.
- **vs. Standard MLP:** Standard networks ignore the "connected" nature of the human body; ST-GCN uses that biological structure as a mathematical advantage.

5. Benefits and Advantages

- **Resource Efficiency:** By using "Lazy Loading" (Data Generators), the model processes data in batches, preventing system crashes.
- **High Accuracy:** The model demonstrated a massive jump from **11% to 75% accuracy** within just two epochs, proving its rapid convergence.
- **Scale Invariance:** Because we normalize coordinates relative to the "Nose" landmark, the model works regardless of how far the user is from the camera.

6. Challenges and Mitigation

The project faced significant engineering hurdles:

- **The "I/O Wall":** Managing 422,073 individual files on a mechanical HDD caused massive latency. We mitigated this by moving to a **Metadata-CSV approach**, which cut preparation time from 20 hours to 5 minutes.
- **Hardware Constraints:** Operating within 8GB of RAM required a **Custom Data Generator** to load only 32 files at a time.
- **Custom Layers:** Implementing the **SpatialGraphConv** layer required custom serialization logic to ensure the model could be saved and reloaded for reporting.

7. Use in Pakistan Sign Language (PSL)

Historically, PSL research has been limited to small datasets and simple CNN models. This project represents a significant step forward as one of the few implementations to apply **Graph Convolutions** to a large-scale PSL dataset. By focusing on 82 distinct classes and nearly half a million samples, this project sets a new benchmark for skeleton-based sign language recognition specifically for the Pakistani context.

8. Conclusion

The ST-GCN represents the "moving forward" choice because it balances academic rigour with practical performance. It successfully transforms the human body into a mathematical graph, allowing for high-speed, high-accuracy recognition that can eventually be deployed on mobile devices for real-time translation, bridging the communication gap for the deaf community in Pakistan.